

Problem Set 10

Taras Halay

28.05.2026

Exercise 3.17

Statement

Exercise 3.17. (Application of orthogonal projection for a minimization problem)

Our goal in this exercise is to approximate a cubic polynomial in a given interval using only polynomials of degree less or equal to 2. Specifically, we wish to find the coefficients of the polynomial $Q(x) = ax^2 + bx + c$ such that the value

$$J[Q] = \int_{-1}^1 [x^3 - Q(x)]^2 dx$$

is minimized.

(i) Let $E = \mathbb{R}_3[X]$ be the real polynomials with degree less or equal to 3. We define the inner product

$$\omega(f, g) = \int_{-1}^1 p(x)q(x) dx.$$

Give the matrix representation $[\omega]_{EE}$, where $E = \{1, x, x^2, x^3\}$ is the standard basis on E .

(ii) Let $R(x) = x^3$, express $J[Q(x)]$ in terms of the norm $\|\cdot\|$ induced by the inner product.

(iii) By definition we know that the polynomial Q belongs to the subspace $F = \mathbb{R}_2[X]$ of real polynomials of degree less or equal to 2. Let $F = \{1, X, X^2\}$ be the standard basis of F , find an orthonormal basis U in F with respect to the inner product $\omega(\cdot, \cdot)$ by applying Gram-Schmidt orthogonalization on the basis E .

(iv) With the orthogonal basis U , apply the result from ex. 2.14 to obtain the projection of R onto the subspace F . Is this polynomial the one that minimizes $J[Q]$?

Solution

(i)

Let $E = \mathbb{R}_3[X]$ be the real polynomials of degree ≤ 3 . Define

$$\omega(p, q) = \int_{-1}^1 p(x)q(x) dx.$$

Let

$$E = \{1, x, x^2, x^3\}$$

be the standard basis of E . Then

$$[\omega]_{EE} = \begin{pmatrix} \omega(1, 1) & \omega(1, x) & \omega(1, x^2) & \omega(1, x^3) \\ \omega(x, 1) & \omega(x, x) & \omega(x, x^2) & \omega(x, x^3) \\ \omega(x^2, 1) & \omega(x^2, x) & \omega(x^2, x^2) & \omega(x^2, x^3) \\ \omega(x^3, 1) & \omega(x^3, x) & \omega(x^3, x^2) & \omega(x^3, x^3) \end{pmatrix}.$$

So

$$([\omega]_{EE})_{i,j} = \omega(x^{i-1}, x^{j-1}) = \int_{-1}^1 x^{i+j-2} dx = \frac{x^{i+j-1}}{i+j-1} \Big|_{-1}^1.$$

The integral is zero if $i+j-1$ is even, i.e. if $i+j$ is odd. If $i+j$ is even, then

$$\int_{-1}^1 x^{i+j-2} dx = \frac{2}{i+j-1}.$$

Therefore

$$[\omega]_{EE} = \begin{pmatrix} 2 & 0 & \frac{2}{3} & 0 \\ 0 & \frac{2}{3} & 0 & \frac{2}{5} \\ \frac{2}{3} & 0 & \frac{2}{5} & 0 \\ 0 & \frac{2}{5} & 0 & \frac{2}{7} \end{pmatrix}.$$

(ii)

Let $R(x) = x^3$. Then

$$J[Q(x)] = \omega(x^3 - Q(x), x^3 - Q(x)) = \|x^3 - Q(x)\|^2.$$

(iii)

$$F = \{1, x, x^2\}.$$

We apply Gram-Schmidt.

$$u_1 = \frac{1}{\|1\|_\omega} = \frac{1}{\sqrt{2}}.$$

Furthermore,

$$u_2 = \frac{x - \omega(x, u_1)u_1}{\|x - \omega(x, u_1)u_1\|_\omega}.$$

Since $\omega(x, u_1) = 0$, we get

$$u_2 = \frac{x}{\|x\|_\omega} = \frac{x}{\sqrt{2/3}} = \frac{\sqrt{3}}{\sqrt{2}}x.$$

Now

$$u_3 = \frac{x^2 - \sum_{i=1}^2 \omega(x^2, u_i)u_i}{\|x^2 - \sum_{i=1}^2 \omega(x^2, u_i)u_i\|_\omega}.$$

We have

$$\omega(x^2, u_1)u_1 = \left(\int_{-1}^1 x^2 \frac{1}{\sqrt{2}} dx \right) \frac{1}{\sqrt{2}} = \frac{1}{3}, \quad \omega(x^2, u_2)u_2 = 0.$$

Thus

$$u_3 = \frac{x^2 - \frac{1}{3}}{\left\|x^2 - \frac{1}{3}\right\|_\omega}.$$

Also

$$\begin{aligned} \left\|x^2 - \frac{1}{3}\right\|_\omega^2 &= \omega\left(x^2 - \frac{1}{3}, x^2 - \frac{1}{3}\right) \\ &= \int_{-1}^1 \left(x^2 - \frac{1}{3}\right)^2 dx \\ &= \int_{-1}^1 \left(x^4 - \frac{2}{3}x^2 + \frac{1}{9}\right) dx = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45}. \end{aligned}$$

Therefore

$$u_3 = \frac{x^2 - \frac{1}{3}}{\sqrt{8/45}}.$$

So

$$U = \left\{ \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{\sqrt{2}}x, \frac{x^2 - \frac{1}{3}}{\sqrt{8/45}} \right\}.$$

(iv)

Since U is normalized,

$$P_F(R) = \sum_{i=1}^3 \langle R, u_i \rangle u_i.$$

We note that

$$\langle x^3, u_1 \rangle = 0, \quad \langle x^3, u_3 \rangle = 0,$$

because odd powers are out of the integral. So

$$P_F(R) = \omega\left(x^3, \frac{\sqrt{3}}{\sqrt{2}}x\right) \frac{\sqrt{3}}{\sqrt{2}}x.$$

Hence

$$P_F(R) = \frac{\sqrt{3}}{\sqrt{2}} \int_{-1}^1 x^4 dx \cdot \frac{\sqrt{3}}{\sqrt{2}}x = \frac{3}{2} \cdot \frac{2}{5}x = \frac{3}{5}x.$$

Let us check minimization:

$$\begin{aligned} \omega\left(x^3 - \frac{3}{5}x, x^3 - \frac{3}{5}x\right) &= \int_{-1}^1 \left(x^3 - \frac{3}{5}x\right)^2 dx \\ &= \int_{-1}^1 \left(x^6 - \frac{6}{5}x^4 + \frac{9}{25}x^2\right) dx = \frac{2}{7} - \frac{6}{5} \cdot \frac{2}{5} + \frac{9}{25} \cdot \frac{2}{3} = \frac{8}{175}. \end{aligned}$$

By this task we reduce our question to

$$\int_{-1}^1 (x^3 - Q(x))^2 dx = \|x^3 - Q\|^2,$$

which is least squares. We need to minimize it, and as it turns out

$$Q = P_F(x^3).$$

Why not using normal distance? Because now

$$\|p - q\|^2 = \int_{-1}^1 (p - q)^2 dx$$

means that if we are close to x^3 on the most of interval, the norm is tiny. If it differs throughout most of interval, the norm is huge.

Why projection? If Q^* is the best approximation in F , then

$$x^3 - Q^* \perp F,$$

which tells

$$\langle x^3 - Q^*, R \rangle = 0, \quad \forall R \in \mathbb{R}_2[X].$$

So, if our $x^3 \in \mathbb{R}_3[X]$, and we are only allowed to approximate it by $Q \in \mathbb{R}_2[X]$, the closest point is where the perpendicular lies. The theorem from the notes says that in any inner-product space, if v is a vector, then the best approximation of v in a subspace F is $\text{proj}_F(v)$.

Exercise 4.7

Statement

Exercise 4.7. (Vandermonde Matrix and Vandermonde Polynomial)

Let $n, m \in \mathbb{N}$ and $x = (x_0, x_1, \dots, x_m) \in \mathbb{K}^{m+1}$, we define the Vandermonde matrix as the $\mathbb{K}^{(m+1) \times (n+1)}$ matrix

$$V(x) = \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \cdots & x_m^n \end{pmatrix},$$

where x_j^k denotes the k -th power of the scalar x_j .

In the following parts we let $n = m$.

- (i) Prove that $\det V(x) = 0$ if and only if $x_i = x_j$ for some $i \neq j$.
- (ii) Using the previous result, show that $\det V(x)$ is divisible by $x_i - x_j$ for all $0 \leq i < j \leq n$.
- (iii) Using the previous result and by comparing the degree of $\prod_{0 \leq i < j \leq m} (x_j - x_i)$ and $\det V(x)$, arrive at the result

$$\det V(x) = C \cdot \prod_{0 \leq i < j \leq m} (x_j - x_i),$$

where $C \in \mathbb{K}$ is some scalar.

(iv) Conclude that $C = 1$, thus

$$\det V(x) = \prod_{0 \leq i < j \leq m} (x_j - x_i).$$

Solution

Let $n, m \in \mathbb{N}$, $x = (x_0, x_1, \dots, x_m) \in \mathbb{K}^{m+1}$. We define the Vandermonde matrix as

$$V(x) = \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \cdots & x_m^n \end{pmatrix}.$$

In the following parts $n = m$.

(i) Prove $\det V(x) = 0$ iff $x_i = x_j$ for some $i \neq j$.

Let us assume that i, j are such that $x_i = x_j$. Then row R_i equals row R_j :

$$R_i = (1, x_i, x_i^2, \dots, x_i^n) = (1, x_j, x_j^2, \dots, x_j^n) = R_j.$$

So $R_i = R_j$ for $i \neq j$. By the alternating property of determinant,

$$\det V(x) = 0.$$

Assume now that $\det V(x) = 0$. Then $\text{rank } V(x) < n + 1$, hence the columns are linearly dependent. Then there exists

$$(a_0, \dots, a_n)^T \neq 0$$

such that

$$V(x) \begin{pmatrix} a_0 \\ \vdots \\ a_n \end{pmatrix} = 0.$$

Then, for $r = 0, \dots, n$,

$$a_0 + a_1 x_r + \cdots + a_n x_r^n = 0.$$

Define

$$p(t) = a_0 + a_1 t + \cdots + a_n t^n.$$

This polynomial is not the zero polynomial, and it has roots $x_0, \dots, x_n$. Since $\deg p \leq n$, it cannot have $n + 1$ distinct roots. Hence

$$x_i = x_j$$

for some $i \neq j$.

(ii) Show that $\det V(x)$ is divisible by $x_i - x_j$ for $0 \leq i < j \leq n$.

Let us focus on x_i, x_j such that $x_i = x_j$. By the previous point,

$$\det V(x) = 0.$$

Consider $\det V(x)$ as a polynomial in the variable x_i . Since it vanishes at $x_i = x_j$, by the factor theorem

$$x_i - x_j \mid \det V(x).$$

Equivalently, up to sign,

$$x_j - x_i \mid \det V(x).$$

Therefore, for every pair (i, j) , the factor $x_j - x_i$ divides $\det V(x)$.

(iii) We now prove the formula for the determinant.

By the Leibniz formula,

$$\det V(x) = \sum_{\sigma \in S_{n+1}} \text{sgn}(\sigma) \prod_{r=0}^n x_r^{\sigma(r)}.$$

So every term has the form

$$x_0^{\sigma(0)} x_1^{\sigma(1)} \cdots x_n^{\sigma(n)}.$$

The degree of each term is

$$\deg \left(\prod_{r=0}^n x_r^{\sigma(r)} \right) = \sum_{r=0}^n \sigma(r) = 0 + 1 + \cdots + n = \frac{n(n+1)}{2}.$$

Since then we just sum up the terms,

$$\deg(\det V(x)) \leq \frac{n(n+1)}{2}.$$

Let us inspect the degree of

$$\prod_{0 \leq i < j \leq n} (x_j - x_i).$$

Since $\deg(x_j - x_i) = 1$, we add up degrees in the product. So we need to count the number of factors, which is exactly the number of pairs (i, j) with $0 \leq i < j \leq n$. This is

$$\binom{n+1}{2} = \frac{n(n+1)}{2}.$$

Hence

$$\deg \left(\prod_{0 \leq i < j \leq n} (x_j - x_i) \right) = \frac{n(n+1)}{2}.$$

From the previous point the product divides $\det V(x)$. Since both degrees are equal, there exists a constant $C \in \mathbb{K}$ such that

$$\det V(x) = C \prod_{0 \leq i < j \leq n} (x_j - x_i).$$

Since we know that $C \in \mathbb{K}$ already, it is sufficient to compare the coefficient of one term. Let us choose

$$x_0^0 x_1^1 \cdots x_n^n.$$

In the Leibniz formula this term comes from the identity permutation. The signature of the identity permutation is 1, so the coefficient of

$$x_0^0 x_1^1 \cdots x_n^n$$

is 1.

In the product

$$\prod_{0 \leq i < j \leq n} (x_j - x_i),$$

choose x_j from every factor. Then we exactly get the term

$$x_0^0 x_1^1 \cdots x_n^n.$$

Since every chosen term has coefficient 1, the coefficient of the whole monomial is also 1. Thus $C = 1$. Therefore

$$\det V(x) = \prod_{0 \leq i < j \leq n} (x_j - x_i).$$

Exercise 5.1

Statement

Exercise 5.1. Let

$$A = \begin{pmatrix} 3 & 0 & -1 \\ 2 & 4 & 2 \\ -1 & 0 & 3 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 0 & -8 \\ 0 & 2 & 0 \\ 4 & 0 & 12 \end{pmatrix}.$$

1. Compute the characteristic polynomial of A and B , and find their eigenvalues.
2. Find the eigensubspaces associated to each eigenvalue.
3. Are the matrices diagonalizable?
4. Compute A^n and B^n for $n \in \mathbb{N}$.

Solution

Let

$$A = \begin{pmatrix} 3 & 0 & -1 \\ 2 & 4 & 2 \\ -1 & 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & -8 \\ 0 & 2 & 0 \\ 4 & 0 & 12 \end{pmatrix}.$$

Matrix A

(i) We compute the characteristic polynomial:

$$A - \lambda I = \begin{pmatrix} 3 - \lambda & 0 & -1 \\ 2 & 4 - \lambda & 2 \\ -1 & 0 & 3 - \lambda \end{pmatrix}.$$

Then

$$\det(A - \lambda I) = (4 - \lambda) \det \begin{pmatrix} 3 - \lambda & -1 \\ -1 & 3 - \lambda \end{pmatrix} = (4 - \lambda)((3 - \lambda)^2 - 1) = (4 - \lambda)^2(2 - \lambda).$$

Hence

$$\lambda_1 = 4, \quad \lambda_2 = 2,$$

and

$$m_4 = 2, \quad m_2 = 1.$$

(ii) For $\lambda = 4$,

$$V_4^A = \ker(A - 4I) = \ker \begin{pmatrix} -1 & 0 & -1 \\ 2 & 0 & 2 \\ -1 & 0 & -1 \end{pmatrix}.$$

The equations give

$$x + z = 0.$$

Therefore

$$V_4^A = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

For $\lambda = 2$,

$$V_2^A = \ker(A - 2I) = \ker \begin{pmatrix} 1 & 0 & -1 \\ 2 & 2 & 2 \\ -1 & 0 & 1 \end{pmatrix}.$$

We get

$$x - z = 0, \quad x + y + z = 0.$$

Thus $x = z$ and $y = -2x$. Therefore

$$V_2^A = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right\}.$$

Here

$$m_2 = d_2 = 1, \quad m_4 = d_4 = 2.$$

Hence A is diagonalizable.

(iii) Let

$$u_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad u_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Then

$$P = (u_1, u_2, u_3) = \begin{pmatrix} 1 & 1 & 0 \\ -2 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

Also

$$P^{-1} = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ 1 & 1 & 1 \end{pmatrix}.$$

Since $A = PDP^{-1}$,

$$A^n = PD^nP^{-1}.$$

Therefore

$$A^n = \begin{pmatrix} \frac{2^n+4^n}{2} & 0 & \frac{2^n-4^n}{2} \\ -2^n+4^n & 4^n & -2^n+4^n \\ \frac{2^n-4^n}{2} & 0 & \frac{2^n+4^n}{2} \end{pmatrix}.$$

Matrix B

(i) From the characteristic polynomial of B,

$$B - \lambda I = \begin{pmatrix} -\lambda & 0 & -8 \\ 0 & 2 - \lambda & 0 \\ 4 & 0 & 12 - \lambda \end{pmatrix}.$$

Thus

$$\det(B - \lambda I) = (2 - \lambda) \det \begin{pmatrix} -\lambda & -8 \\ 4 & 12 - \lambda \end{pmatrix} = (2 - \lambda)(\lambda^2 - 12\lambda + 32) = (2 - \lambda)(\lambda - 4)(\lambda - 8).$$

So

$$\lambda_1 = 2, \quad \lambda_2 = 4, \quad \lambda_3 = 8.$$

(ii) For $\lambda = 2$,

$$V_2^B = \ker(B - 2I) = \ker \begin{pmatrix} -2 & 0 & -8 \\ 0 & 0 & 0 \\ 4 & 0 & 10 \end{pmatrix}.$$

The equations give

$$x = 0, \quad z = 0,$$

so

$$V_2^B = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

For $\lambda = 4$,

$$V_4^B = \ker(B - 4I) = \ker \begin{pmatrix} -4 & 0 & -8 \\ 0 & -2 & 0 \\ 4 & 0 & 8 \end{pmatrix}.$$

Therefore

$$x + 2z = 0, \quad y = 0.$$

Thus

$$V_4^B = \text{span} \left\{ \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

For $\lambda = 8$,

$$V_8^B = \ker(B - 8I) = \ker \begin{pmatrix} -8 & 0 & -8 \\ 0 & -6 & 0 \\ 4 & 0 & 4 \end{pmatrix}.$$

We get

$$x + z = 0, \quad y = 0.$$

Hence

$$V_8^B = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}.$$

All algebraic multiplicities and each eigenvector geometric multiplicity are equal to 1, hence B is diagonalizable.

(iii) To compute B^n , form the matrix

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

and

$$P = \begin{pmatrix} 0 & -2 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}.$$

Then

$$P^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & -1 \\ -1 & 0 & -2 \end{pmatrix}.$$

Since

$$B = PDP^{-1},$$

we get

$$B^n = PD^nP^{-1}.$$

Therefore

$$B^n = \begin{pmatrix} 0 & -2 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 2^n & 0 & 0 \\ 0 & 4^n & 0 \\ 0 & 0 & 8^n \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & -1 \\ -1 & 0 & -2 \end{pmatrix}.$$

Since middle terms cancel,

$$B^n = \begin{pmatrix} 2 \cdot 4^n - 8^n & 0 & 2 \cdot 4^n - 2 \cdot 8^n \\ 0 & 2^n & 0 \\ 8^n - 4^n & 0 & 2 \cdot 8^n - 4^n \end{pmatrix}.$$

Exercise 5.2

Statement

For $n \in \mathbb{N}$, $\mathbb{K} = \mathbb{C}$, determine the eigenvalues and eigenspaces of

$$A = \begin{pmatrix} i & i & 0 & 0 & \cdots & 0 & 0 \\ 0 & i & i & 0 & \cdots & 0 & 0 \\ 0 & 0 & i & i & \cdots & 0 & 0 \\ 0 & 0 & 0 & i & \ddots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & i & i \\ 0 & 0 & 0 & 0 & \cdots & 0 & i \end{pmatrix}, \quad i = \sqrt{-1}.$$

Solution

Since the matrix is upper triangular,

$$\det(A - \lambda I) = (i - \lambda)^n.$$

Thus

$$\lambda = i$$

is the only eigenvalue. Now

$$V_i^A = \ker(A - iI).$$

But

$$A - iI = \begin{pmatrix} 0 & i & 0 & \cdots & 0 & 0 \\ 0 & 0 & i & \cdots & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & i & 0 \\ 0 & 0 & 0 & \cdots & 0 & i \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} 0 & i & 0 & \cdots & 0 & 0 \\ 0 & 0 & i & \cdots & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & i & 0 \\ 0 & 0 & 0 & \cdots & 0 & i \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = 0.$$

This gives

$$ix_j = 0 \Rightarrow x_j = 0, \quad j = 2, \dots, n.$$

Hence

$$V_i^A = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \right\}.$$

So

$$\dim(V_i^A) = 1 = d_i,$$

while

$$m_i = n.$$

Hence A is diagonalizable iff $n = 1$.